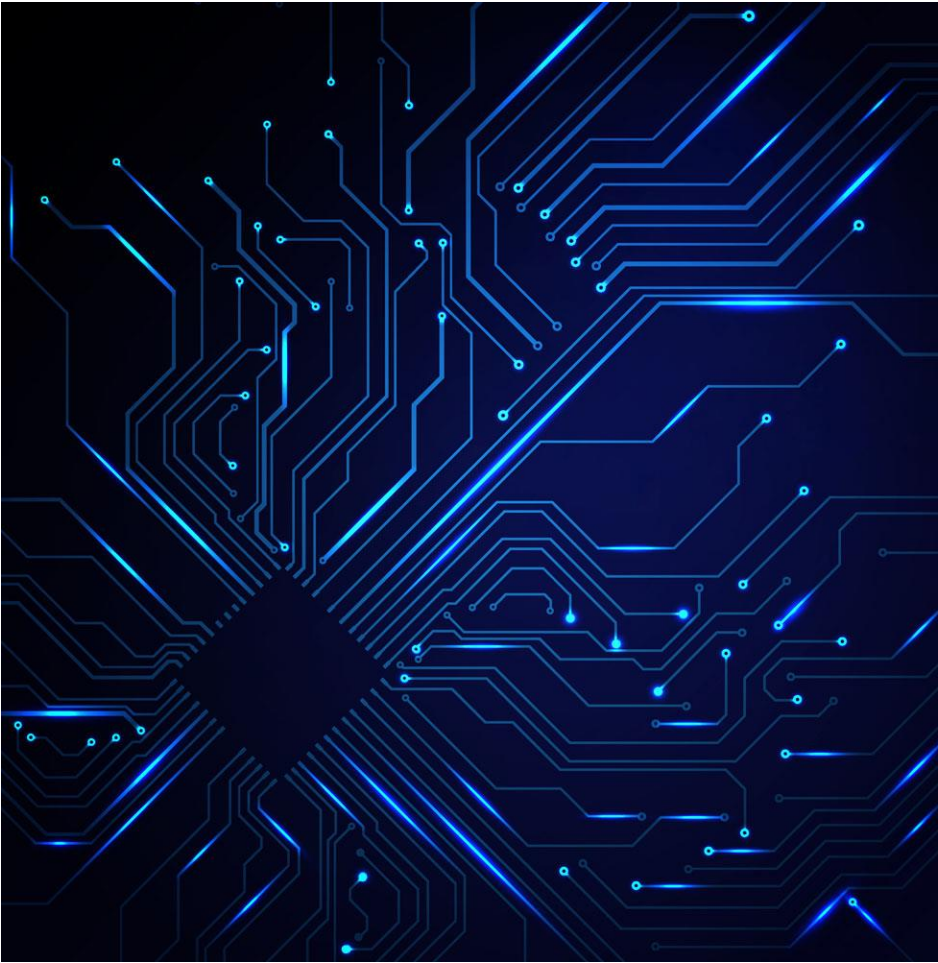




Notice

1. Lab report must be submitted in PDF format.
2. Lab report must be submitted to the TronClass by 13:00 next class hour.
3. Each student must submit one lab report.
4. File name of the lab report must be in Lab#_StudentID.

LAB 2-1

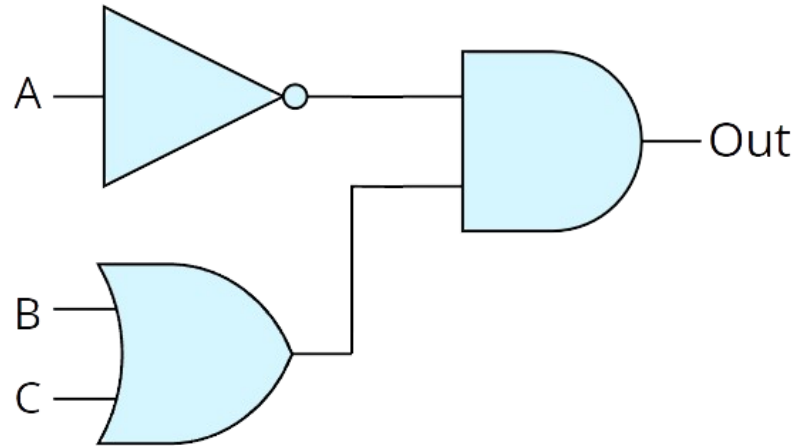
BASIC LOGIC GATE IMPLEMENTATION – PART 2

Group : 14

Name : 黃翊宏

Student ID : 01357101

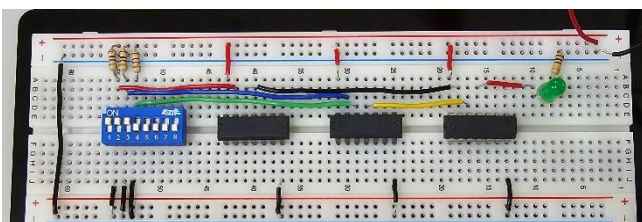
● Lab 2-1



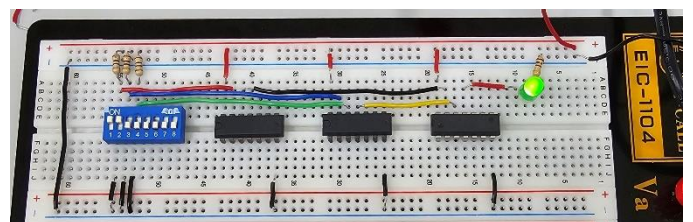
Truth Table (Voltage)				Truth Table (Logic)			
A	B	C	out	A	B	C	out
0	0	0	0.17	0	0	0	0
0	0	1	2.98	0	0	1	1
0	1	0	2.98	0	1	0	1
0	1	1	2.98	0	1	1	1
1	0	0	0.17	1	0	0	0
1	0	1	0.17	1	0	1	0
1	1	0	0.17	1	1	0	0
1	1	1	0.17	1	1	1	0

Breadboard Wiring Photos
(Output must be connected to an LED and clearly labeled)

A = 0 ; B = 0 ; C = 0

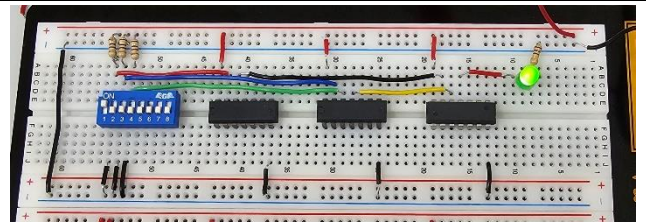
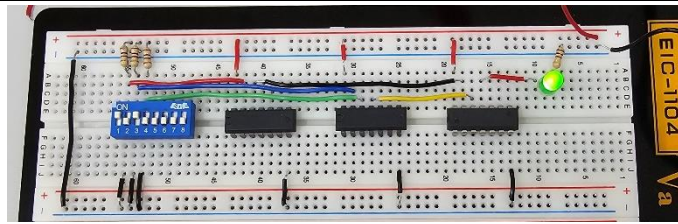


A = 0 ; B = 0 ; C = 1



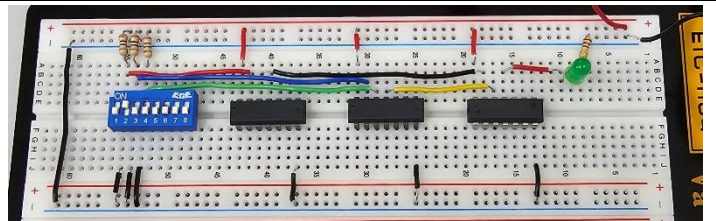
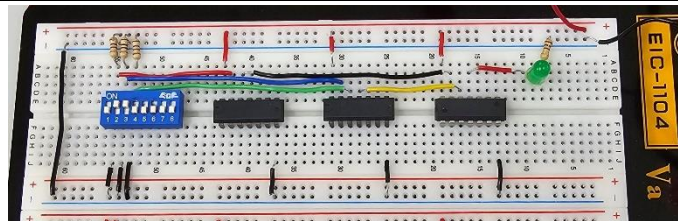
$A = 0 ; B = 1 ; C = 0$

$A = 0 ; B = 1 ; C = 1$



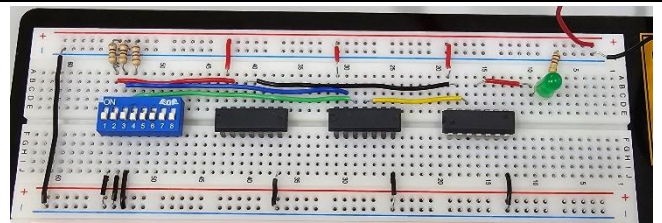
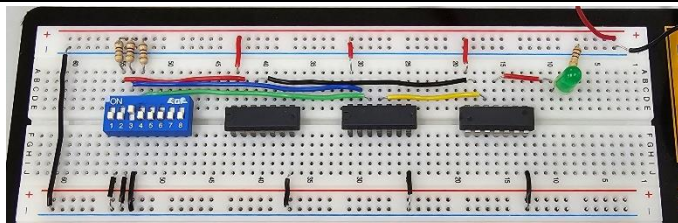
$A = 1 ; B = 0 ; C = 0$

$A = 1 ; B = 0 ; C = 1$

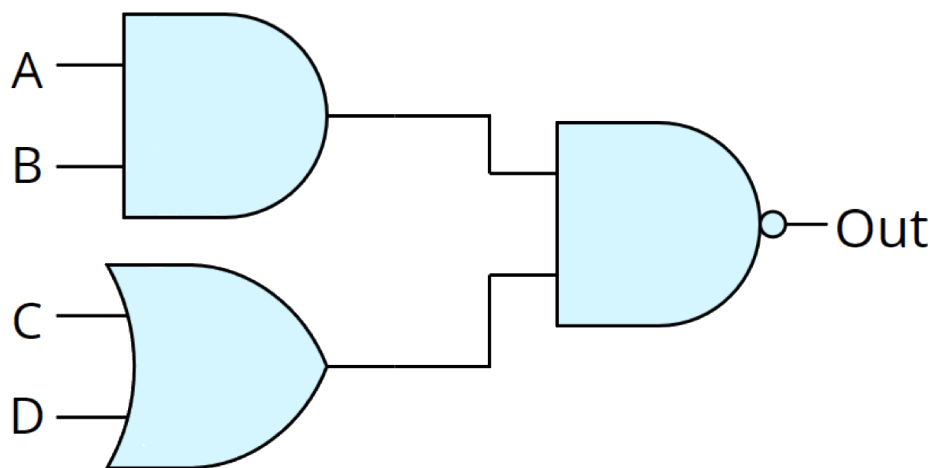


$A = 1 ; B = 1 ; C = 0$

$A = 1 ; B = 1 ; C = 1$



● Lab 2-2

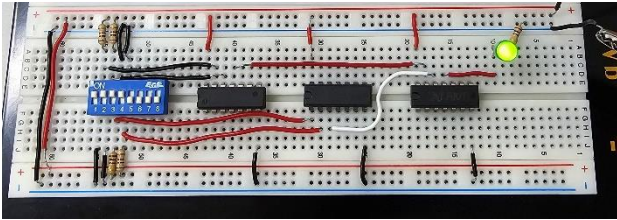


Truth Table (Voltage)					Truth Table (Logic)				
A	B	C	D	out	A	B	C	D	out
0	0	0	0	2.92	0	0	0	0	1
0	0	0	1	2.92	0	0	0	1	1
0	0	1	0	2.92	0	0	1	0	1
0	0	1	1	2.92	0	0	1	1	1
0	1	0	0	2.92	0	1	0	0	1
0	1	0	1	2.92	0	1	0	1	1
0	1	1	0	2.92	0	1	1	0	1
0	1	1	1	2.92	0	1	1	1	1
1	0	0	0	2.92	1	0	0	0	1
1	0	0	1	2.92	1	0	0	1	1
1	0	1	0	2.92	1	0	1	0	1
1	0	1	1	2.92	1	0	1	1	1
1	1	0	0	2.92	1	1	0	0	1
1	1	0	1	0.12	1	1	0	1	0

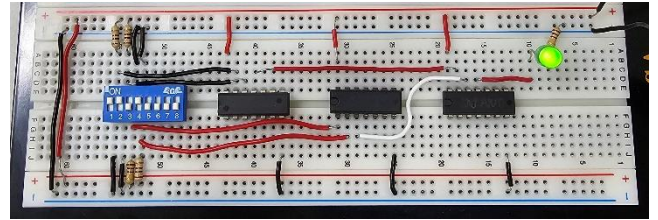
1	1	1	0	0.12	1	1	1	0	0
1	1	1	1	0.12	1	1	1	1	0

Breadboard Wiring Photos
(Output must be connected to an LED and clearly labeled)

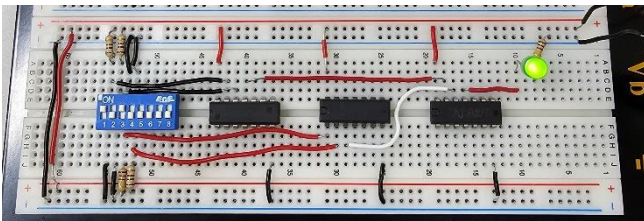
$A = 0 ; B = 0 ; C = 0 ; D = 0$



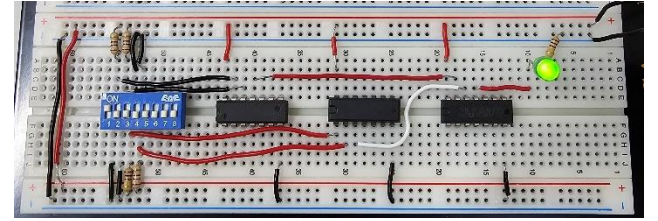
$A = 0 ; B = 0 ; C = 1 ; D = 0$



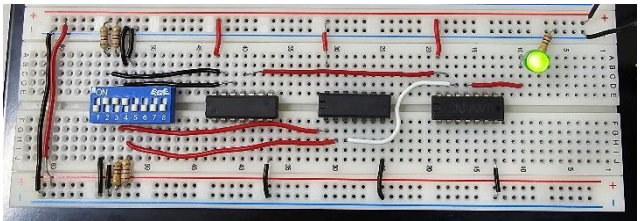
$A = 0 ; B = 1 ; C = 0 ; D = 1$



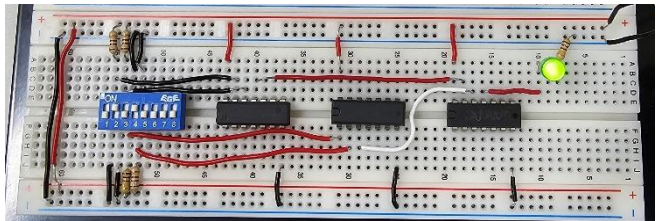
$A = 0 ; B = 1 ; C = 1 ; D = 1$



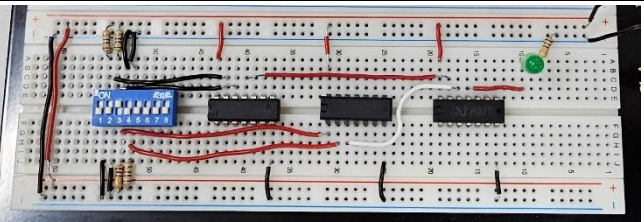
$A = 1 ; B = 0 ; C = 0 ; D = 0$



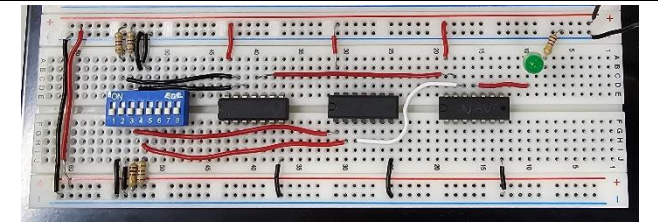
$A = 1 ; B = 0 ; C = 1 ; D = 0$



$A = 1 ; B = 1 ; C = 0 ; D = 1$



$A = 1 ; B = 1 ; C = 1 ; D = 1$



Lab Summary and Reflection :

Summary:

In this experiment, we continued our study of TTL IC applications by implementing logic circuits using four different TTL ICs: NAND (7400), OR (7432), AND (7408), and NOT (7404). The objective was to construct circuits, verify their corresponding truth tables, and analyze their outputs. By following the given circuit diagrams, we successfully implemented and tested each logic gate, ensuring the expected logical behaviors were achieved. Additionally, we measured voltage levels and compared them with theoretical logic values to confirm circuit accuracy. Photos were taken to document the experiment setup and results.

Reflection:

Compared to the previous experiment, we optimized our workflow by improving task delegation, which allowed us to complete the experiment more efficiently. Furthermore, after our last experiment, we spent extra time sorting and categorizing ICs and resistors, which helped prevent issues caused by faulty components in this session. This preparation ensured a smoother experiment process with minimal troubleshooting. Overall, the experiment was highly successful, reinforcing our understanding of TTL logic gates and practical circuit implementation.